Anytime-Valid Runtime Monitoring of Learned Barrier Certificates

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Abstract

Learned barrier certificates provide a principled mechanism for establishing safety of stochastic systems, but their validity may deteriorate during deployment when the underlying dynamics differ from those used for training. We propose an anytime-valid runtime monitor that audits a learned barrier certificate using observed closed-loop trajectories. The monitor is intended to detect loss of statistical support for the deployed barrier condition; it does not by itself guarantee detection before unsafe entry or safety of a subsequent intervention. Specifically, the conditional barrier-drift condition is formulated as an episode-wise sequential hypothesis test, and observed barrier increments are transformed into a nonnegative test supermartingale, yielding finite-sample false-alarm control under arbitrary stopping times. Inspired by finite-horizon safety verification, we further introduce a deterministic temporal shaping of the monitoring boundary, making the monitor more conservative early in a prescribed deployment horizon while preserving anytime-validity. The monitor is designed to detect statistical evidence that the safety certificate is no longer consistent with the observed dynamics, rather than to predict unsafe entry. To assess the monitor empirically, we evaluate empirical alarm behavior under valid and invalid certificates, compare betting strategies, and examine the trade-off between unnecessary alarms and detection timeliness.

1 Introduction

Safety is a critical requirement for learning-enabled autonomous systems operating in uncertain and evolving environments. In robotics, autonomous driving, and cyber-physical infrastructure, even rare failures can be catastrophic, making purely data-driven validation insufficient for deployment (Chougule et al. 2023; Uesato et al. 2018). Consider an autonomous drone navigating a cluttered warehouse whose floor plan changes over time. A barrier trained in simulation may provide a probabilistic safety guarantee under the assumed system dynamics and disturbance model, but unmodeled disturbances or environmental changes may invalidate the deployed safety certificate. This motivates runtime monitoring mechanisms that continuously assess whether the assumptions supporting a deployed safety certificate remain consistent with observed system behavior and, when necessary, trigger appropriate safety interventions.

Barrier certificates (Prajna and Jadbabaie 2004; Prajna, Jadbabaie, and Pappas 2007) certify safety through a scalar function $B(x)$ whose expected value does not increase along trajectories, offering a scalable alternative to reachability analysis. In modern practice, however, the barrier B is learned from data, which introduces a fundamental verification gap: finite datasets provide incomplete coverage, simulation-to-reality mismatch induces distribution shift, and black-box models resist formal verification. In stochastic environments, iterative test-then-train refinement on counterexamples also cannot guarantee global validity of the updated certificate.

Rather than pursuing complete verification, we shift the focus to *runtime monitoring* and *alarm-triggered intervention*. The key idea is to continuously audit the deployed barrier using observed closed-loop trajectories and detect statistical evidence that the certificate is no longer consistent with the underlying dynamics. When sufficient evidence accumulates, the monitor can trigger a predefined intervention or fallback mechanism, thereby providing a practical mechanism for responding to potential loss of certificate validity during deployment. It is worth noting that the monitor is not designed to predict the exact time of unsafe entry, nor does it guarantee detection before every safety violation. Its purpose is to identify statistically significant evidence that the validity of the safety certificate is in question.

To realize this, we formulate runtime barrier auditing as an anytime-valid sequential hypothesis test based on e-processes (test supermartingales) and predictable plug-in betting strategies (Ramdas et al. 2023; Shafer et al. 2011; Vovk and Wang 2021; Shafer 2021; Waudby-Smith and Ramdas 2022). Observed barrier-value increments are converted into betting factors whose product forms a nonnegative test supermartingale. Consequently, the monitor retains finite-sample false-alarm control under arbitrary stopping times within an episode, allowing evidence to be assessed continuously without sacrificing the prescribed statistical validity. A finite deployment horizon also provides useful temporal structure for monitoring. Inspired by finite-horizon safety verification (Xue 2025), we introduce a deterministic time-varying scaling of the monitoring boundary that makes the monitor more conservative at early time steps and gradually approaches the standard anytime-valid threshold as the prescribed horizon is reached. The resulting framework combines anytime-valid runtime monitoring with a finite-horizon temporal boundary tailored to the deployment horizon.

The formal guarantee concerns the statistical validity of the barrier-condition monitor: when the learned barrier satisfies its conditional supermartingale condition, the probability of an alarm within an episode is controlled at the prescribed level. The theory does not guarantee detection of every barrier violation, detection before unsafe entry, or safety of the intervention taken after an alarm. These properties are therefore evaluated empirically and distinguished from the statistical false-alarm guarantee.

The contributions of this work are summarized as follows.

1. **Anytime-valid barrier auditing:** We formulate the conditional drift condition underlying a stochastic barrier certificate as an episode-wise sequential hypothesis test and construct an e-process from observed barrier increments. The monitor provides finite-sample level-α false-alarm control under arbitrary within-episode stopping times.
2. **Finite-horizon boundary shaping:** Motivated by finite-horizon safety verification results (Xue 2025), we introduce a deterministic time-varying scaling of the monitoring boundary to tailor the monitor to a prescribed finite deployment horizon. The resulting exponential boundary profile is more conservative at early time steps and gradually approaches the standard anytime-valid threshold as the deployment horizon is reached.

Related Work

1). Runtime Verification and Statistical Monitoring. Runtime Verification (RV) monitors executions against formal specifications such as linear temporal logic (LTL) or signal temporal logic (STL) (Bauer, Leucker, and Schallhart 2011), but struggles in stochastic environments where deviations may arise from transient noise. Statistical Model Checking (SMC) (Clarke and Zuliani 2011) addresses stochastic systems through fixed-horizon sampling, but lacks time-uniform guarantees and support for arbitrary stopping times. Recent predictive RV approaches combine conformal prediction with STL specifications (Lindemann et al. 2023; Zhao et al. 2024; Cairoli, Paoletti, and Bortolussi 2023) and adaptive conformal inference (Gibbs and Candès 2021) to maintain coverage under distribution shifts. However, these methods provide only marginal or asymptotic guarantees and cannot ensure anytime-valid false-alarm control. Several works incorporate runtime monitoring into safety assurance. The adaptive shielding approach of (Feng et al. 2025) refines verified safe control envelopes online, but requires an expert-specified dynamics model and a verified fallback policy. Similarly, the shielding methods of (Alshiekh et al. 2018; Jansen et al. 2020) assume a known system model (deterministic or MDP) for offline safety verification. Relatedly, (Yu, Žikelić, and Henzinger 2025) develops runtime monitors for detecting or predicting violations of learned certificates in deterministic continuous-time systems and uses the collected data for offline repair. In contrast, our framework targets stochastic systems, where barrier conditions are defined through conditional expectations, and formulates runtime monitoring as an anytime-valid sequential hypothesis test based on e-processes (Ramdas et al. 2023; Shafer et al. 2011; Vovk and Wang 2021). The method is model-free, relying only on online observed trajectories, and provides evidence-triggered runtime intervention with uniform false-alarm control under arbitrary stopping times within each episode.

2). Barrier Certificates and Synthesis. Barrier certificates have been widely studied for safety verification and controller synthesis (Prajna and Jadbabaie 2004; Jagtap, Soudjani, and Zamani 2018; Cao et al. 2025; Xue 2025; Ames et al. 2019; Žikelić et al. 2023; Zhi et al. 2024; Abate, Giacobbe, and Roy 2024; Xue 2024; Henzinger et al. 2025; Xue 2026). Existing data-driven methods (Mathiesen et al. 2023; Nejati and Zamani 2023; Salamati et al. 2024; Wu et al. 2026; Mathiesen et al. 2026) focus on synthesizing barrier certificates under assumptions such as known dynamics. In contrast, our work does not synthesize barrier certificates; rather, it provides an anytime-valid runtime monitor for auditing deployed barrier certificates and detecting statistical evidence that their underlying drift conditions may no longer be consistent with the observed dynamics, particularly under distribution shifts. The resulting alarm-triggered trajectories can be reused as counterexamples for subsequent certificate updating using existing synthesis methods. This creates a complementary relationship with CEGIS (Abate et al. 2021; Zhao et al. 2023; Chatterjee et al. 2023): whereas CEGIS focuses on iterative certificate synthesis and refinement, our framework focuses on deployment-time monitoring of a learned certificate.

Paper organization. The remainder of the paper is organized as follows. Section 2 presents the problem formulation. Section 3 develops the proposed anytime-valid runtime monitoring framework, including the e-process construction, adaptive betting strategy, and finite-horizon boundary shaping. Section 4 evaluates the monitor on stochastic benchmark systems under both valid and invalid barrier certificates. Section 5 concludes the paper and discusses future directions.

2 Preliminaries

This section introduces the mathematical setting underlying our framework. We first review e-processes and anytime-valid inference, and then formalize the stochastic control system and the runtime safety monitoring problem.

2.1 E-Processes and Anytime-Valid Inference

We work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, where Ω is the sample space, $\mathcal{F}$ is a σ-algebra, and $\mathbb{P}$ is a probability measure. Random variables are measurable functions defined on this probability space.

Definition 1 (Filtration)**.** *A filtration is an increasing sequence of sub-σ-algebras $\{\mathcal{F}_t\}_{t\geq 0}$ of $\mathcal{F}$ such that $\mathcal{F}_s \subseteq \mathcal{F}_t$ for all $s \leq t$.*

A filtration models the information available at each time t; $\mathcal{F}_t$ contains exactly what is known up to that point.

Definition 2 (Martingale and supermartingale)**.** *Let $(M_t)_{t\geq 0}$ be a sequence of random variables adapted to $(\mathcal{F}_t)_{t\geq 0}$. We say that $(M_t)_{t\geq 0}$ is a* martingale *if*

1. *M_t is $\mathcal{F}_t$-measurable for all $t \geq 0$,*
2. *$\mathbb{E}[|M_t|] < \infty$ for all $t \geq 0$,*
3. *$\mathbb{E}[M_{t+1} \mid \mathcal{F}_t] = M_t$ almost surely for all $t \geq 0$.*

If the equality $\mathbb{E}[M_{t+1} \mid \mathcal{F}_t] = M_t$ *is weakened to* $\mathbb{E}[M_{t+1} \mid \mathcal{F}_t] \leq M_t$*, then* (M_t) *is called a* supermartingale*. In this case the process drifts downward in conditional expectation. Every martingale is a supermartingale.*

When the filtration is understood, we simply refer to $(M_t)_{t\geq 0}$ as a martingale or supermartingale.

Definition 3 (E-process and test supermartingale)**.** *A nonnegative adapted process* $(M_t)_{t\geq 0}$ *is called an* e-process *under a null hypothesis* H_0 *if* $\mathbb{E}_{H_0}[M_\tau] \leq 1$ *for every bounded stopping time* τ*. A* test supermartingale *is an e-process that is also a supermartingale with* $M_0 = 1$*. Every test supermartingale is an e-process, but the converse is not true.*

The power of these objects comes from a time-uniform tail bound. While Markov's inequality controls a single nonnegative random variable, Ville's inequality extends the control to the whole trajectory of a test supermartingale.

Proposition 1 (Ville's inequality (Ville 1939))**.** *Let* $(M_t)_{t\geq 0}$ *be a nonnegative test supermartingale under* H_0*. Then for any* $\alpha \in (0, 1)$*,* $\mathbb{P}_{H_0}\left(\sup_{t\geq 0} M_t \geq \frac{1}{\alpha}\right) \leq \alpha$*.*

If the process ever crosses the threshold $1/\alpha$, we may reject H_0 with a false-alarm probability at most α.

A standard method to construct a test supermartingale is through *e-variables*. Let $(E_t)_{t\geq 1}$ be a sequence of nonnegative random variables such that E_t is $\mathcal{F}_t$-measurable and $\mathbb{E}_{H_0}[E_t \mid \mathcal{F}_{t-1}] \leq 1$. Then the product process $M_t := \prod_{k=1}^{t} E_k$ with $M_0 := 1$ is a nonnegative supermartingale and hence an e-process. Each factor E_k can be seen as a local bet against the null; the product aggregates this evidence multiplicatively. Persistent small violations of the null accumulate steadily, while isolated fluctuations have little long-term effect. By Ville's inequality, the test that stops and rejects H_0 the first time $M_t \geq 1/\alpha$ controls the type-I error at level α, uniformly over all stopping times.

2.2 Problem Formulation

We consider a discrete-time stochastic dynamical system

$$x_{k+1} = f(x_k, u_k, \omega_k), \tag{1}$$

where $x_k \in \mathbb{R}^n$ is the state, $u_k \in \mathcal{U}$ is the control input, and ω_k denotes an independent and identically distributed (i.i.d.) stochastic disturbance. *Throughout this paper, both the system dynamics* f *and the disturbance distribution are assumed to be unknown.* The system is operated under a feedback policy $\pi : \mathbb{R}^n \to \mathcal{U}$, yielding the closed-loop dynamics $u_k = \pi(x_k)$. Let $(\mathcal{F}_k)_{k\geq 0}$ denote the natural filtration generated by the trajectory.

Safety objective. Let $\mathcal{S} \subseteq \mathbb{R}^n$ denotes a safe set, $\mathcal{X}_0 \subset \mathcal{S}$ denotes the initial set. The safety objective is to ensure that trajectories starting from an initial state $x_0 \in \mathcal{X}_0$ remain within $\mathcal{S}$ with high probability, i.e., $\mathbb{P}(x_k \in \mathcal{S},\ \forall k \geq 0)$.

A standard approach to certifying such properties is through *barrier certificates*. To formalize the analysis, it is convenient to introduce the stopped (or frozen) system

$$\hat{x}_{k+1} = \begin{cases} f(\hat{x}_k, \pi(\hat{x}_k), \omega_k), & \text{if } \hat{x}_k \in \mathcal{S}, \\ \hat{x}_k, & \text{otherwise.} \end{cases}$$

This construction ensures that once the trajectory exits $\mathcal{S}$, it remains outside. The probability that the original system stays within $\mathcal{S}$ coincides with the probability that the stopped system never leaves $\mathcal{S}$ (Yu et al. 2023).

Barrier certificate condition (Kushner 1967; Yu et al. 2023). Assume that the barrier function $B : \mathbb{R}^n \to \mathbb{R}_{\geq 0}$ is integrable such that the following conditional expectation is well-defined. The function B is called a barrier certificate if it satisfies

$$B(x) \geq 0, \qquad \forall x \in \mathbb{R}^n, \tag{2a}$$
$$B(x) \geq 1, \qquad \forall x \notin \mathcal{S}, \tag{2b}$$
$$\mathbb{E}\left[B(\hat{x}_{k+1}) \mid \hat{x}_k = x\right] \leq B(x), \qquad \forall x \in \mathbb{R}^n. \tag{2c}$$

The last condition states that B behaves as a supermartingale along the stopped system. If such a function exists, then for any $x_0 \in \mathcal{X}_0$, $\mathbb{P}(x_k \in \mathcal{S},\ \forall k \geq 0) \geq 1 - B(x_0)$.

Remark 1. *In this paper, we consider the barrier condition of the form* (2c) *for safety verification. Nevertheless, the proposed method is not restricted to this specific formulation and can be extended to more general barrier conditions, such as* $\mathbb{E}\left[B(\hat{x}_{k+1}) \mid \hat{x}_k = x\right] \leq aB(x) + b, \forall x \in \mathbb{R}^n$*, which is widely used in finite-time and infinite-time safety verification, as well as reach-avoid analysis (Xue 2025, 2026).*

Problem of interest. We consider the setting where both the barrier certificate B and the controller π are *learned from data*, so condition (2c) may fail globally despite appearing sound on the training data. Rather than assuming the correctness of these learned components, our goal is to continuously audit their validity from realized trajectories while maintaining a time-uniform false-alarm probability bounded by a prescribed level α. The resulting monitor can serve as a component of a runtime safety-shielding mechanism: upon detecting statistical evidence that the barrier certificate is no longer trusted, it triggers a predefined conservative intervention to reduce the risk of subsequent unsafe behavior.

3 Anytime-Valid Runtime Monitoring of Learned Barrier Certificates

In this section, we introduce our anytime-valid sequential monitoring framework for runtime auditing of learned barrier certificates (Fig. 1). Instead of requiring offline validation of a deployed barrier certificate, we continuously audit its consistency with observed closed-loop trajectories. By formulating barrier-condition monitoring as a sequential hypothesis test and constructing an e-process that accumulates evidence online, we obtain an anytime-valid monitor with finite-sample false-alarm control under arbitrary stopping times within an episode. The resulting evidence can be used to trigger predefined runtime interventions when the validity of the deployed safety certificate is called into question.

3.1 Anytime-Valid Monitoring

Trajectories and Barrier Consistency Hypothesis. *We first formalize the observed closed-loop trajectories and the barrier consistency hypothesis that the monitor will audit*

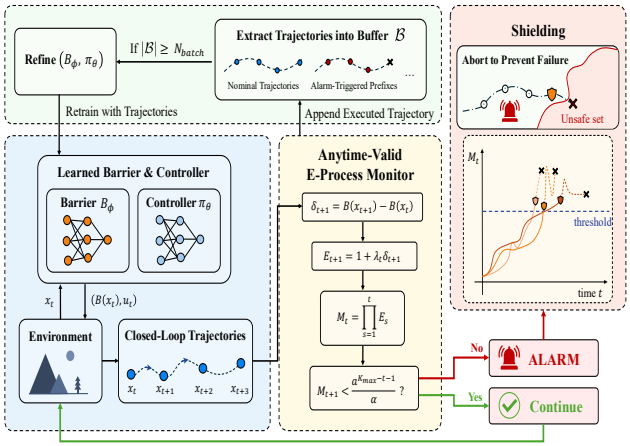


Figure 1: Overview of the proposed framework for anytime-valid monitoring of learned barrier certificates.

throughout deployment. A single run of the closed-loop system under the learned controller π produces a trajectory

$$\xi := \{\hat{x}_0, \hat{x}_1, \hat{x}_2, \dots\},$$

with natural filtration $\mathcal{F}_t := \sigma(\hat{x}_0, \dots, \hat{x}_t)$, which captures all information available up to time t within that specific episode, where $\hat{x}_{i+1} = f(\hat{x}_i, \pi(\hat{x}_i), w_i)$, $i = 0, 1, 2, \dots$.

The first two conditions of the barrier certificate B can be constructed alone. Our task is to test whether the conditional supermartingale condition (2c) remains valid under the deployed dynamics. The monitor uses one realized trajectory to accumulate statistical evidence against this conditional null. The null hypothesis is therefore

$$H_0: \quad \mathbb{E}\Big[B(\hat{x}_{t+1}) \mid \mathcal{F}_t\Big] \leq B(\hat{x}_t), \tag{3}$$

for all $t \geq 0$. Equivalently, defining the one-step increment

$$\delta_{t+1} := B(\hat{x}_{t+1}) - B(\hat{x}_t),$$

we test whether $\mathbb{E}[\delta_{t+1} \mid \mathcal{F}_t] \leq 0$. Under the null the barrier should not drift upward in expectation; any systematic tendency to increase signals a potential safety violation.

Local Evidence. We now convert each observed increment into an evidence factor that behaves like a fair bet against the null. Define

$$E_{t+1} := 1 + \lambda_t \delta_{t+1},$$

where the *betting fraction* $\lambda_t \geq 0$ is chosen at time t using only information available up to that point in the same episode ($\mathcal{F}_t$-measurable). Intuitively, if the barrier increases ($\delta_{t+1} > 0$), the factor exceeds 1, signalling a surprise; if the barrier decreases, the factor is ≤ 1, consistent with H_0.

Proposition 2 (One-step validity)**.** *Under H_0, if $\lambda_t \geq 0$, then $\mathbb{E}_{H_0}\Big[E_{t+1} \mid \mathcal{F}_t\Big] \leq 1$.*

Proof. By definition, $\mathbb{E}_{H_0}\Big[E_{t+1} \mid \mathcal{F}_t\Big] = 1 + \lambda_t \mathbb{E}_{H_0}\Big[\delta_{t+1} \mid \mathcal{F}_t\Big]$. Under H_0 the conditional expectation of δ_{t+1} is non-positive, and $\lambda_t \geq 0$, we have the conclusion. □

Thus, under the null, each factor E_{t+1} has conditional expectation at most 1 – it is a "fair bet" in the sense that we cannot expect to accumulate evidence against a true null. The betting fraction λ_t controls the sensitivity of the monitor: larger values react more strongly to positive increments but require a tighter bound to preserve non-negativity.

Evidence Accumulation. Local bets are combined multiplicatively into a single cumulative score that reflects the total evidence against the null accumulated during the episode. The trajectory-level evidence process is

$$M_t := \prod_{s=1}^{t} E_s, \qquad M_0 := 1.$$

Because each factor E_s has conditional expectation ≤ 1, the product is a non-negative supermartingale under H_0, a fact we record as a lemma if every factor E_s satisfies $E_s \geq 0$.

Lemma 1. *Under H_0 and $E_s \geq 0$, $\forall s \in \mathbb{N}$, $(M_t)_{t \geq 0}$ is a test supermartingale.*

Proof. Non-negativity is obvious. The process is adapted, and $\mathbb{E}_{H_0}[M_{t+1} \mid \mathcal{F}_t] = M_t \, \mathbb{E}_{H_0}[E_{t+1} \mid \mathcal{F}_t] \leq M_t$, where we used that M_t is $\mathcal{F}_t$-measurable and the one-step bound from Proposition 2. □

The product form of the e-process enables sustained evidence against the barrier condition to accumulate multiplicatively over time. When the observed barrier increments consistently provide evidence against H_0, the corresponding betting factors can compound the evidence substantially, whereas transient fluctuations whose effects do not persist have less lasting influence on the accumulated score. This allows the monitor to respond to systematic departures from the learned barrier condition without treating every stochastic fluctuation as a certificate violation.

Because $\{M_t\}_{t \geq 0}$ is a test supermartingale, Ville's inequality applied to the stopping rule $\tau := \inf\{t \geq 1 : M_t \geq 1/\alpha\}$ bounds the false-alarm probability of any episode by α: if the evidence process crosses the threshold, we reject H_0, and under H_0 this occurs with probability at most α. The threshold crossing should therefore be interpreted as evidence that the deployed barrier condition has lost statistical support, rather than as a direct prediction of unsafe entry. The following subsection exploits a prescribed finite deployment horizon to introduce a time-varying monitoring boundary.

3.2 Data-driven adaptive betting on λ_t (DAB)

Even if the above theory is established, we still encounter the following two difficulties: 1. How to choose a suitable λ_t that can satisfy the nonnegativity constraint on E_s in Lemma 1; 2. Even if the constraints are met, how to adapt these bets to the observed barrier increments? Below we will introduce the design method of λ_t based on data-driven and using discounted empirical moments inspired by Adam.

Building on the Kelly betting principle in (Waudby-Smith and Ramdas 2022; Shafer 2021), we use an exponentially weighted version of the empirical first and second moments, inspired by the moment estimates in Adam (Kingma and Ba 2014). The unweighted case is recovered when $\gamma = 1$.

Specifically, we choose λ_t based only on barrier increments observed up to time t, $\delta_s = B(\hat{x}_s) - B(\hat{x}_{s-1})$ by setting $S_t = \gamma S_{t-1} + \delta_t, Q_t = \gamma Q_{t-1} + \delta_t^2$, and

$$\lambda_t^{\mathrm{K}} = \max\left(\frac{S_t}{Q_t + \varepsilon}, 0\right), \qquad \varepsilon > 0$$

where $\gamma \in (0, 1]$ is a fixed memory parameter. Both moments contain only increments observed by time t, before observing δ_{t+1}; we set $S_0 = Q_0 = \lambda_0 = 0$, and $\varepsilon = 1 \times 10^{-12}$, and then use

$$\lambda_t = \min\left\{\lambda_t^{\mathrm{K}},\, U,\, -\frac{1}{\underline{\delta}_t}\right\} \geq 0. \tag{4}$$

Here, $U > 0$ prevents excessively large bets and $\underline{\delta}_t = -B(\hat{x}_t)$. Then $\delta_{t+1} = B(\hat{x}_{t+1}) - B(\hat{x}_t) \geq -B(\hat{x}_t) = \underline{\delta}_t$, leading to the conservative cap $\lambda_t \leq 1/B(\hat{x}_t)$. Under this choice, $E_{t+1} = 1 + \lambda_t \delta_{t+1} \geq 1 - 1 \geq 0$.

The intuition follows a growth-rate and Kelly-type principle. When the observed barrier increments fluctuate around zero, the estimated conditional drift remains small, resulting in a conservative betting fraction. In contrast, when the observed increments provide sustained evidence of a positive barrier drift, the estimated drift increases λ_t, allowing the e-process to accumulate evidence more rapidly. Thus, the adaptive betting rule places greater weight on persistent departures from the learned barrier condition while remaining conservative when the observed dynamics provide little evidence against it.

We introduce γ to address a specific cancellation effect in barrier increments: without forgetting ($\gamma = 1$), $S_t = \sum_{s=1}^{t} \delta_s = B(\hat{x}_t) - B(\hat{x}_0)$, so a sustained recent increase can remain masked by an earlier decrease, whereas $\gamma < 1$ discounts stale increments and allows the betting fraction to respond to increases relative to a recent barrier baseline. In the appendix, we provide a detailed demonstration of the scenarios in which the $\gamma = 1$ case fails.

Remark 2. *When only the nonnegativity of the barrier function is known, the bound $\underline{\delta}_t = -B(\hat{x}_t)$ can be overly conservative, as it relies on the worst-case barrier decrease and may lead to slow evidence accumulation. In practice, tighter empirical bounds $\underline{\delta}_t$ can be estimated from offline rollouts. These data-driven bounds may improve detection sensitivity but do not provide worst-case guarantees for unseen deployment conditions. Accordingly, we use $\underline{\delta}_t = -B(\hat{x}_t)$ as the default choice throughout the paper, and consider empirical bounds only as non-guaranteed alternatives in the Appendix.*

3.3 Shaping the Abort Time via Deterministic Scaling

While the constant-threshold rule remains valid under arbitrary stopping times, it does not exploit the finite deployment horizon $K_{\max}$ that often arises from practical constraints such as physical timeouts, fuel limits, or fixed simulation lengths. A finite horizon allows the monitoring boundary to be shaped so that early alarms require stronger evidence, reducing sensitivity to transient fluctuations, while the threshold gradually approaches the standard anytime-valid level as the horizon is reached. We therefore introduce a deterministic temporal scaling of the boundary that adjusts the monitor's conservatism over the episode while preserving the false-alarm guarantee α.

We can achieve this temporal preference by scaling the supermartingale with a deterministic time factor, effectively shaping the abort boundary without altering the underlying evidence factor E_s:

$$M_t^a = a^t M_t, \qquad a \geq 1.$$

Under H_0, because $\mathbb{E}_{H_0}[M_{t+1} \mid \mathcal{F}_t] \leq M_t$, the conditional expectation of the scaled process satisfies:

$$\mathbb{E}_{H_0}[M_{t+1}^a \mid \mathcal{F}_t] = a^{t+1}\mathbb{E}_{H_0}[M_{t+1} \mid \mathcal{F}_t] \leq a^{t+1} M_t = a M_t^a.$$

Thus, the expected value of M_{t+1}^a grows by at most a factor of a. Following the finite-horizon martingale properties established in Theorem 1 in (Xue 2025) (with $M_0^a = 1$), the probability of crossing the terminal threshold is bounded by α. We can formalize this by defining the sequential stopping rule as the first moment the scaled monitor crosses $\frac{a^{K_{\max}}}{\alpha}$:

$$\tau := \inf\left\{t \leq K_{\max} : M_t^a \geq \frac{a^{K_{\max}}}{\alpha}\right\},$$

which guarantees that $\mathbb{P}_{H_0}(\tau \leq K_{\max}) \leq \alpha$. Using $M_t^a = a^t M_t$, the stopping rule can be equivalently written as:

$$\tau := \inf\left\{t \leq K_{\max} : M_t \geq \frac{a^{K_{\max} - t}}{\alpha}\right\}.$$

Therefore, the time-varying threshold preserves the desired false-alarm control while allowing earlier detection to be emphasized through the parameter a. For $a = 1$, the rule reduces to the constant-threshold baseline. If $\tau \leq K_{\max}$, the trajectory is stored as a counterexample and the monitor triggers a predefined intervention intended to reduce the risk of subsequent unsafe behavior.

Theorem 1 (Anytime-valid falsification guarantee)**.** *Let H_0 be as in* (3)*. Suppose that $(M_t)_{t \geq 0}$ is a nonnegative test supermartingale under H_0. Define the rejection event $\mathcal{R} := \{\tau \leq K_{\max}\}$. Then $\mathbb{P}_{H_0}(\mathcal{R}) \leq \alpha$.*

Proof. Define the scaled process

$$V_t := \frac{\alpha}{a^{K_{\max}}} M_t^a.$$

Since $(M_t)_{t \geq 0}$ is a nonnegative test supermartingale under H_0,

$$\begin{aligned}
\mathbb{E}_{H_0}[V_{t+1} \mid \mathcal{F}_t] &= \frac{\alpha}{a^{K_{\max}}} a^{t+1} \mathbb{E}_{H_0}[M_{t+1} \mid \mathcal{F}_t] \\
&\leq \frac{\alpha}{a^{K_{\max}}} a^{t+1} M_t \\
&= a V_t.
\end{aligned}$$

Thus, V_t satisfies the finite-horizon growth condition

$$\mathbb{E}_{H_0}[V_{t+1} \mid \mathcal{F}_t] \leq \frac{V_t}{\bar{\alpha}} + \bar{\beta}$$

from Theorem 1 in (Xue 2025), with

$$\bar{\alpha} = \frac{1}{a} \in (0, 1], \qquad \bar{\beta} = 0.$$

Moreover,

$$V_0 = \frac{\alpha}{a^{K_{\max}}}, \qquad \bar{\gamma} = \bar{\beta}\bar{\alpha} - (\bar{\alpha} - 1) = 1 - \frac{1}{a} \in [0, 1].$$

Therefore, the case $\bar{\alpha} \in (0, 1]$ and $\bar{\gamma} \in [0, 1]$ of Theorem 1 in (Xue 2025) gives

$$\begin{aligned}\mathbb{P}_{H_0}\left(\sup_{0\le t\le K_{\max}} V_t \ge 1\right) &\le V_0 \bar{\alpha}^{-K_{\max}} \\ &= \frac{\alpha}{a^{K_{\max}}} a^{K_{\max}} \\ &= \alpha.\end{aligned}$$

Finally,

$$V_t \ge 1 \quad \Longleftrightarrow \quad M_t^a \ge \frac{a^{K_{\max}}}{\alpha} \quad \Longleftrightarrow \quad M_t \ge \frac{a^{K_{\max}-t}}{\alpha}.$$

Hence,

$$\left\{\sup_{0\le t\le K_{\max}} V_t \ge 1\right\} = \{\tau \le K_{\max}\},$$

and therefore

$$\mathbb{P}_{H_0}(\tau \le K_{\max}) \le \alpha.$$

This completes the proof. □

Ultimately, this deterministic scaling mechanism has profound consequences for the temporal dynamics of the abort time.

- **Case** $a = 1$**:** $M_t^a = M_t$ is a test supermartingale; the rejection threshold is $1/\alpha$ and the test is valid for any stopping time (no bias toward early or late abort). While theoretically clean, it ignores the finite horizon and treats all time steps equally, which can be wasteful when resources are limited or early decisions are preferred.
- **Case** $a > 1$ **(patient test):** $a^{K_{\max}-t}$ decreases exponentially over time, making the rejection threshold $\frac{a^{K_{\max}-t}}{\alpha}$ highest initially and lowest near the deadline. Thus, early alarms require stronger evidence, while the test approaches the classical anytime-valid rule as $t \to K_{\max}$. This design is preferable when false alarms are costly or transient effects are common. For example, a drone may experience short-lived oscillations after takeoff; a patient monitor avoids unnecessary aborts by requiring sustained evidence before intervention. Larger values of a improve robustness against transient fluctuations and spurious alarms, while values closer to 1 recover the standard monitor. This conservatism, however, reduces sensitivity to brief or rapidly developing safety violations.

By framing a as a patience parameter, the safety engineer can deliberately prioritize robustness against early noise or rapid detection and resource efficiency, choosing the mode that best aligns with the system's physical characteristics and the cost of data collection.

3.4 Comparison with Direct Barrier Monitoring

Since the barrier value $B(\hat{x}_t)$ is a supermartingale under H_0, one may ask whether monitoring $\frac{B(\hat{x}_t)}{B(\hat{x}_0)}$ is sufficient, making the proposed e-process unnecessary. Indeed, Ville's inequality guarantees the same episode-wise false-alarm control by stopping whenever $\frac{B(\hat{x}_t)}{B(\hat{x}_0)} \ge \frac{1}{\alpha}$. Although statistically valid, direct barrier monitoring suffers from several practical limitations that the multiplicative e-process avoids.

1. **Universal stopping thresholds.** Direct monitoring depends on the initial barrier value $B(\hat{x}_0)$, so the stopping threshold varies across episodes. In contrast, the e-process always starts from $M_0 = 1$, yielding the universal threshold $1/\alpha$ independent of the initial state.
2. **Greater sensitivity to persistent violations.** The e-process accumulates evidence multiplicatively, allowing small but systematic deviations from the barrier condition to compound rapidly. As a result, it detects gradual certificate degradation substantially earlier than monitoring the barrier value alone.
3. **Flexible temporal shaping.** In practical deployments, monitoring is limited to a finite horizon $K_{\max}$. The proposed e-process admits the deterministic temporal scaling factor a, enabling a principled trade-off between early detection and robustness to transient fluctuations while preserving the prescribed false-alarm guarantee. By contrast, direct barrier monitoring offers no comparable mechanism for adjusting its temporal operating characteristics.

Remark 3. *Choosing* $\lambda_t = \frac{1}{B_t}$ *gives* $E_{t+1} = 1 + \lambda_t \delta_{t+1} = \frac{B_{t+1}}{B_t}$*, and therefore* $M_t = \prod_{s=1}^{t} E_s = \frac{B_t}{B_0}$ *with* $M_0 = 1$*. Thus, the e-process reduces exactly to the normalized barrier process.*

3.5 Runtime Monitoring

The preceding subsections establish the statistical foundations of the proposed anytime-valid monitor. We now instantiate these results as a deployment-time runtime monitoring procedure.

In each deployment episode, the learned barrier B_ϕ and policy π_θ are fixed, and the monitor updates the e-process online using the observed closed-loop trajectory. The episode is terminated when the evidence process crosses the time-varying decision boundary. Under the null hypothesis that the deployed barrier satisfies the conditional drift condition, the probability of such an alarm within an episode is controlled at the prescribed level α, regardless of the stopping time. The monitoring procedure is independent of how the barrier is constructed and can therefore be applied to analytical, optimization-based, or learned barrier certificates, provided that the assumptions required by the monitor are satisfied. The collected trajectories can optionally be stored as counterexamples and used for subsequent barrier or policy refinement between episodes; such updates do not occur within an episode and are therefore separate from the anytime-valid monitoring procedure. The complete deployment procedure is summarized in Algorithm 1.

Algorithm 1: Anytime-Valid Runtime Monitoring

Require: Initial barrier B_0, policy π_0, buffer size N_{batch}, significance level α, maximum horizon $K_{\max}$, and patience factor $a \geq 1$
Ensure: Runtime safety interventions
Initialize $B_\phi \leftarrow B_0$, $\pi_\theta \leftarrow \pi_0$, $\mathcal{B} \leftarrow \emptyset$
for deployment episode $r = 1, 2, \ldots$ **do**
  Observe $\hat{x}_0$; initialize $M_0 \leftarrow 1$ and $\xi \leftarrow (\hat{x}_0)$
  for $t = 0, 1, \ldots, K_{\max} - 1$ **do**
    Choose $\lambda_t \geq 0$ as in Eq. (4)
    Set $u_t \leftarrow \pi_\theta(\hat{x}_t)$ and observe $\hat{x}_{t+1}$
    Append $(\hat{x}_t, u_t, \hat{x}_{t+1})$ to ξ
    Compute $\delta_{t+1} \leftarrow B_\phi(\hat{x}_{t+1}) - B_\phi(\hat{x}_t)$
    Update $M_{t+1} \leftarrow M_t(1 + \lambda_t \delta_{t+1})$
    if $M_{t+1} \geq a^{K_{\max}-(t+1)}/\alpha$ **then**
      Exit the episode; **break**
    end if
  end for
  $\mathcal{B} \leftarrow \mathcal{B} \cup \{\xi\}$
  if $|\mathcal{B}| \geq N_{\text{batch}}$ **then**
    Optional: Update B_ϕ, π_θ using $\mathcal{B}$; set $\mathcal{B} \leftarrow \emptyset$
  end if
end for

Remark 4. *The monitoring procedure is intended to operate while the system remains in the safe set $\mathcal{S}$. By construction, the stopped process follows the original system's dynamics whenever it is in the safe set $\mathcal{S}$ and remains frozen after exiting $\mathcal{S}$. Consequently, the global supermartingale condition*

$$\mathbb{E}\left[B(\hat{\boldsymbol{x}}_{k+1}) \mid \hat{\boldsymbol{x}}_k = \boldsymbol{x}\right] \leq B(\boldsymbol{x}), \qquad \forall \boldsymbol{x} \in \mathbb{R}^n,$$

is equivalent to the barrier-drift condition for the original system restricted to the safe set:

$$\mathbb{E}\left[B(\boldsymbol{x}_{k+1}) \mid \boldsymbol{x}_k = \boldsymbol{x}\right] \leq B(\boldsymbol{x}), \qquad \forall \boldsymbol{x} \in \mathcal{S}.$$

Indeed, outside $\mathcal{S}$, the stopped process remains at its current state, so the supermartingale condition holds automatically with equality. Thus, the safe set $\mathcal{S}$ is precisely the region in which the barrier-drift condition needs to be monitored. Accordingly, each trajectory is terminated upon its first exit from $\mathcal{S}$ or upon reaching the prescribed horizon $K_{\max}$, whichever occurs first. Thus, once a trajectory exits the safe set, there is no need to simulate its subsequent evolution.

Remark 5. *The monitor is designed to detect statistical evidence against the learned barrier condition, rather than to predict unsafe entry. At the same time, the monitor can serve as a component of a runtime safety-shielding mechanism by triggering a predefined conservative intervention when sufficient evidence indicates that the learned certificate is no longer consistent with the observed dynamics. Thus, a trajectory may move toward the unsafe region without triggering the monitor, provided that the monitored barrier condition remains statistically consistent with the observed dynamics. Such a situation does not constitute a failure of the proposed detection objective, since the monitor is intended to identify when the statistical support for the safety certificate has become questionable, rather than to predict every impending safety violation. Once sufficient evidence against the barrier condition accumulates, the deployed certificate is treated as unsupported for continued operation, and the predefined intervention can be activated to reduce the risk of subsequent unsafe behavior.*

Alg. 1 resets the e-process at the start of each episode, yielding false-alarm control at level $\alpha_r = \alpha$ within each episode. The per-episode guarantee is appropriate when episodes are treated as separate operational units, as in episodic RL or session-based monitoring. If error control is instead required across N_{ep} deployment episodes, the episode-wise error budget can be allocated such that

$$\sum_{r=1}^{N_{\text{ep}}} \alpha_r \leq \alpha,$$

for example, using $\alpha_r = \frac{\alpha}{N_{\text{ep}}}$ or a summable sequence such as $\alpha_r = \frac{\alpha}{2^r}$. By the union bound, the probability of at least one false alarm over all N_{ep} episodes is then at most α. More elaborate cross-episode e-process constructions may also be considered when their required measurability and null conditions are satisfied. Such global error control involves a sensitivity trade-off: decreasing α_r across episodes increases the rejection threshold $\frac{1}{\alpha_r}$ and may reduce detection power in later episodes. We therefore use per-episode error control in our experiments and leave cross-episode error allocation as an application-dependent extension.

4 Experiments

We evaluate the proposed anytime-valid runtime monitoring framework on four benchmark systems. The experiments investigate three questions:

1. **RQ1:** whether the monitor preserves the prescribed false-alarm guarantee when the barrier certificate is valid;
2. **RQ2:** Can we detect environmental changes and invalid barrier certificates using our monitor?
3. **RQ3:** How does temporal boundary shaping trade fewer false alarms under valid certificates for delayed or missed detection under invalid certificates?

The default setting is $K_{\max} = 400$, $K_{\max} \ln a = 4$, $\gamma = 0.92$, and $U = 100$, using the same runs. Table 1 also reports the standard anytime-valid rule $K_{\max} \ln a = 0 (a = 1)$. Subsection 4.1 answers RQ1, Subsection 4.2 answers RQ2. Subsection 4.3 answers RQ3. Appendix B.3 analyzes the sensitivity of $K_{\max}$ and λ_t. Detailed benchmark descriptions, implementation details, proofs of validity or counterexamples establishing invalidity for the candidate barriers are provided in Appendix A. We also compare with other monitors besides the direct betting: Fraction mixture, absolute mixture and CUSUM in Appendix B.

Why do we use $K_{\max} \ln a$? Since a enters the alarm threshold exponentially over the horizon, varying a directly makes it difficult to interpret the strength of threshold shaping. We therefore parameterize a in terms of $K_{\max} \ln a$, which directly determines the logarithm of the threshold inflation factor at the beginning of the horizon relative to the terminal threshold. With this parameterization, the alarm threshold at step k is

$$\ln M_k \;\geq\; \ln(1/\alpha) + K_{\max} \ln a \left(1 - \frac{k}{K_{\max}}\right).$$

Thus, at $k = 0$, the threshold lies $K_{\max} \ln a$ above $\ln(1/\alpha)$, and it decreases linearly in k to $\ln(1/\alpha)$ at $k = K_{\max}$. The threshold $a^{K_{\max}-k}/\alpha$ on M_k starts at $e^{K_{\max} \ln a}/\alpha$ and decreases exponentially to $1/\alpha$.

The formal objective of the proposed framework is runtime monitoring of the learned barrier condition. Theorem 1 provides a statistical guarantee for this objective: under H_0, the probability of falsely rejecting the barrier-drift condition within an episode is at most α. This guarantee does not imply a lower bound on detection power or guarantee that an alarm occurs before an unsafe transition. Nevertheless, the monitor can serve as a component of a runtime safety-shielding mechanism: when sufficient evidence is accumulated against the deployed barrier, a predefined intervention can be triggered rather than continuing to rely on an unsupported barrier. We therefore evaluate unsafe-rollout detection as a downstream operational detection measure of this monitoring mechanism in Subsection 4.4.

4.1 Runtime Monitoring under Valid Certificates

Under a valid certificate, Theorem 1 guarantees that the probability of an alarm within the horizon $K_{\max}$ is at most α for the system starting from the state in the initial set $\mathcal{X}_0$. To empirically assess the consistency of the observed alarm behavior within this theoretical guarantee, we sample N initial states $x_0^{(1)}, \ldots, x_0^{(N)}$ and simulate R independent trajectories from each initial state. Let $\tau^{(i,j)}$ be the alarm time of the monitor on the j-th trajectory from $x_0^{(i)}$. The empirical alarm rate for $x_0^{(i)}$ is defined as

$$\hat{p}_i = \frac{1}{R} \sum_{j=1}^{R} \mathbb{I}\big(\tau^{(i,j)} <= K_{\max}\big).$$

as well as the worst-case empirical alarm rate,

$$\hat{p}^{\max} = \max_i \hat{p}_i.$$

To summarize the empirical alarm behavior across the sampled initial states, we call $x_0^{(i)}$ empirically abnormal if $\hat{p}_i > \alpha$ and report the fraction of empirically abnormal initial states,

$$\rho_\alpha = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\big(\hat{p}_i > \alpha\big),$$

Under a valid certificate, we expect ρ_α to be close to zero and $\hat{p}^{\max}$ to remain close to or below α, providing empirical support for the theoretical guarantee. Conversely, it is worth noting that observing empirical alarm rates below α does not, by itself, establish the validity of the certificate which will be discussed in detail in the Appendix B.3.

System	$K_{\max} \ln a$	Direct ρ_α	Direct $\hat{p}^{\max}$	DAB ρ_α	DAB $\hat{p}^{\max}$
Valid certificates					
System 1	0	0.00	4.80	0.00	0.90
	4	0.00	0.20	0.00	0.10
System 2	0	0.00	0.00	0.00	0.00
	4	0.00	0.00	0.00	0.00
Invalid certificates					
System 2-input	0	18.10	100.00	82.10	92.00
	4	17.90	100.00	79.90	81.50
System 3	0	24.00	69.40	79.00	96.30
	4	23.80	69.40	78.40	95.10

Table 1: Runtime monitoring under valid and invalid certificates, using the standard rule ($K_{\max} \ln a = 0$) and the default setting ($K_{\max} \ln a = 4$) on the same runs. ρ_α is the fraction of initial states with $\hat{p}_i > \alpha$ and $\hat{p}^{\max} = \max_i \hat{p}_i$ is the worst-case alarm rate. $N = 10^3$ initial states, $R = 10^3$ trajectories per initial state, $K_{\max} = 400$, $\alpha = 0.05$. All values are percentages.

Results. We evaluate the monitor on two benchmark systems, for which globally valid barrier certificates satisfying (2) are available. We also compare two betting strategies: DAB and Direct ($\lambda_t = \frac{1}{B(\hat{x}_t)}$). As shown in Table 1, both monitors achieve $\hat{p}^{\max} < \alpha$ under both rules, which is consistent with the conclusion of $\mathbb{P}_{H_0}(\mathcal{R}) \leq \alpha$ in Theorem 1. However, the rates for the direct betting remain slightly higher than those for the DAB. As shown in Figure 5, System 1 has lower values in the initial points $B(x_0)$, thereby increasing the probability of an alarm when $B_t/B_0 > 1/\alpha$. In contrast, our method is far less sensitive to the initial value and is therefore much less affected by this issue. Overall, these observations are both consistent with the anytime-valid false-alarm guarantee under a valid certificate.

4.2 Runtime Monitoring under Invalid Certificates

In this subsection, we will answer RQ2 by investigating scenarios involving invalid barriers. The remaining systems in Table 1 have invalid barrier certificates. We consider two scenarios: (1) initially, we obtain a valid barrier function for System 2 in RQ1. However, environmental changes are inevitable in real-world scenarios. To simulate such a scenario, we actively injected noise b_k into state x_2 from step $k = 100$ onward:

$$\begin{aligned} x_{2,k+1} = x_{2,k} + 0.01\big(&(0.8 + \xi_k)\, x_{1,k} \\ &+ 10\,(x_{1,k}^2 - 0.21)\, x_{2,k} + b_k\big) \end{aligned} \tag{5}$$

where $b_k \sim \mathcal{U}[0, 0.45]$ and the equation of x_1 is unchanged. Although the true environmental dynamics parameters were

unknown, we tested the original valid function against the modified dynamical system and discovered counterexamples. Consequently, the original barrier function proved to be invalid in the changed environment. (2) The environment has not changed, but due to the sample-based training method, the barrier function for system 3 is invalid from the start. We also find a counterexample establishing that this candidate is invalid.

Results. When $K_{\max} \ln a = 4$, under invalid certificates, both methods detect substantial deviations from the prescribed alarm behavior, but their performance differs across the two systems. In System 2-input attack, direct betting has a worst-case alarm rate of 100%, compared with 81.5% for DAB. However, DAB identifies a much larger fraction of abnormal initial states, with $\rho_\alpha = 79.9\%$ compared with 17.9% for direct betting. In System 3, DAB performs better in terms of both metrics: its ρ_α reaches 78.40%, compared with 23.80% for direct betting, while its worst-case alarm rate is 95.10%, compared with 69.40% for direct betting. These results highlight the complementary roles of the two metrics. A high worst-case alarm rate $\hat{p}^{\max}$ indicates that the monitor frequently raises alarms for at least one sampled initial state, suggesting that the deployed barrier condition may no longer be trustworthy. Meanwhile, ρ_α quantifies the fraction of sampled initial states for which the empirical alarm rate exceeds α, revealing how widely this evidence of potential barrier invalidity occurs across the initial states.

4.3 The patience parameter a

We introduce a as a deployment-time design parameter for the monitoring boundary. For fixed α, $K_{\max}$, and betting strategy, changing a leaves the underlying evidence process M_k unchanged and modifies only the decision rule applied to that evidence. Its choice expresses the trade-off between avoiding unnecessary intervention and detecting certificate invalidity promptly. A larger a is appropriate when unnecessary early interventions are costly and the application can tolerate delayed detection.

The experimental results in Subsections 4.1 and 4.2, reported in Table 1 and Figure 5, show that increasing a cannot increase the number of alarms and cannot advance the first alarm time. This observation is consistent with the theoretical analysis: a larger a raises the rejection threshold at early time steps, making the monitor more conservative. To illustrate how this affects monitoring under valid and invalid certificates, we consider Systems 1 and System2+input. In System 1, where the barrier certificate is valid, the DAB evidence process exhibits a transient increase that triggers an alarm when $a = 1$, but then drops rapidly (we give an example trajectory in Figure 2). Since the certificate is valid in this scenario, this threshold crossing constitutes a false alarm which is caused by a temporary fluctuation rather than sustained evidence against the barrier condition. Increasing the temporal scaling to $K_{\max} \ln a = 4$ raises the early-time rejection threshold, preventing the evidence process from triggering an alarm. In this case, the more conservative threshold helps avoid an unnecessary alarm. In contrast, in System 2+input, where the barrier certificate is invalid, increasing the temporal scaling to $K_{\max} \ln a = 4$ delays the alarm compared with the case $a = 1$, but the evidence eventually becomes sufficient to cross the rejection threshold. Thus, the monitor still detects evidence against the barrier condition, albeit later. These examples illustrate the trade-off controlled by a: a more conservative threshold can suppress alarms caused by transient fluctuations under a valid certificate, but may also delay the detection of an invalid certificate.

In practice, before deployment, we do not know whether this barrier is valid. Thus, a should be selected according to the desired level of confidence in the deployed certificate and the trade-off between avoiding unnecessary alarms and detecting degradation promptly. When the certificate is considered highly trustworthy and avoiding unnecessary alarms is the priority, a larger a can be used to make the monitor more conservative. In contrast, when the validity of the certificate is uncertain or degradation is suspected, setting $a = 1$ is preferable because it allows earlier detection. In particular, we recommend $a = 1$ when prompt detection of potential certificate invalidity is the primary objective. When there is some disturbance in the early stages of the system, we recommend choosing $a > 1$.

4.4 Practical Application of Algorithm 1

A large number of alarms indicates that the deployed barrier certificate can no longer be trusted. Following Algorithm 1, we then update it using data collected from the environment: for System 2 under the input attack we retrain only the barrier function, and for System 3 we retrain both the barrier function and the policy. We then check whether the alarm rates return below α.

Results. For System 2 under the input attack, retraining the barrier function reduces the worst-case alarm rate $\hat{p}^{\max}$ over all initial states from 100.00% to 0.00% for the direct rule and from 81.50% to 0.00% for DAB. For System 3, $\hat{p}^{\max}$ decreases from 69.40% to 0.00% for the direct rule and from 95.10% to 0.00% for DAB. After retraining, neither monitor raises an alarm in any run of either system, so $\rho_\alpha = 0$. Under a valid certificate, Theorem 1 bounds the alarm probability by α. The observed alarm rates are consistent with this, and the monitors find no evidence against the retrained certificates. With the accumulation of evidence M_k, we find that our method can terminate the trajectory in advance, before the state leaves the safe set $\mathcal{S}$. In the system 2, there is no trajectory leading away from S after the environment changes. But for the system 3 in the 42.88% of trajectories left $\mathcal{S}$, the DAB betting achieves an early termination rate of 47.09% prior to leaving $\mathcal{S}$, whereas the direct betting method achieves 11.88% (with $a = 1$: 51.81% and 22.56%).

5 Conclusions and Future Work

We introduced an anytime-valid framework for runtime monitoring of learned barrier certificates in stochastic systems. By continuously auditing observed barrier-value increments with an e-process, the proposed monitor provides finite-sample false-alarm control under arbitrary stopping times within each deployment episode. Motivated by finite-horizon safety verification, we further introduced a temporal boundary-shaping mechanism that adjusts the conservatism

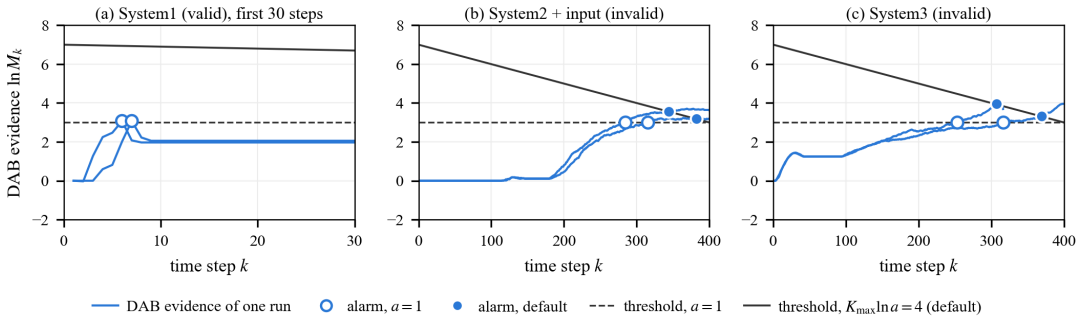


Figure 2: The trade-off between avoiding unnecessary alarms and detecting degradation promptly.

of the monitor over a prescribed deployment horizon. Experiments under valid certificates, certificate degradation, and distribution shifts demonstrate the practical utility of the framework for detecting statistical evidence against the barrier condition and supporting conservative runtime intervention.

Several directions remain for future work. First, extending the framework to safe RL, continuous-time dynamics, and partially observable systems would broaden its applicability. Second, validation on real robotic platforms under non-stationary environments would further demonstrate its practical value. Third, the relationship between barrier-condition violations and detection power, detection delay, and pre-violation detection warrants further theoretical and empirical study. Moreover, the finite-horizon factor a can be further studied as a time-varying parameter to adapt the monitoring strategy.

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A Benchmark details

A function B is a barrier certificate if it satisfies the three conditions (2a), (2b), and (2c). For a state $x \in \mathcal{S}$, let x^+ denote its next state. For $x \notin \mathcal{S}$, the stopped system stays at x, so (2c) holds with equality there, and it only has to be checked on $\mathcal{S}$. To show that a candidate is valid, all three conditions must hold everywhere. To show that it is invalid, we can just find a single *counterexample* $x \in \mathcal{S}$ suffices: $\mathbb{E}[B(x^+) \mid x] > B(x)$.

System1: stochastic polynomial system (1D). *Dynamics.* The state $x \in \mathbb{R}$ evolves according to

$$x_{k+1} = (0.5 + 1.5\xi_k)x_k(1 - x_k^2), \qquad \xi_k \sim \mathcal{U}[-1, 1].$$

The disturbances are independent across time.
Sets and sampling.

$$\mathcal{X}_0 = [-0.4, 0.4], \;\; \mathcal{S} = (-1, 1).$$

Initial states are sampled uniformly from $\mathcal{X}_0$, independently of the disturbance sequence.

System 1: valid. We choose the candidate $B(x) = x^2$. It satisfies (2a), and (2b) holds since $|x| \geq 1$ implies $x^2 \geq 1$. For (2c), $x^+ = (0.5 + 1.5\xi)\, x(1 - x^2)$ with $\xi \sim \mathcal{U}[-1, 1]$, and $\mathbb{E}[(0.5 + 1.5\xi)^2] = 0.25 + 1.5\,\mathbb{E}[\xi] + 2.25\,\mathbb{E}[\xi^2] = 0.25 + 0 + 0.75 = 1$. Hence, for every $x \in \mathcal{S}$,

$$\begin{aligned}\mathbb{E}[B(x^+) \mid x] - B(x) &= x^2(1 - x^2)^2 - x^2 \\ &= -x^4(2 - x^2) \leq 0.\end{aligned}$$

System2: Van der Pol oscillator (2D) (Wu et al. 2024). *Dynamics.* The state $x = (x_1, x_2) \in \mathbb{R}^2$ evolves:

$$\begin{aligned}x_{1,k+1} &= x_{1,k} + 0.01\,(-2x_{2,k}), \\ x_{2,k+1} &= x_{2,k} + 0.01\big((0.8 + \xi_k)\, x_{1,k} \\ &\qquad + 10\,(x_{1,k}^2 - 0.21)\, x_{2,k}\big),\end{aligned}$$

where $\xi_k \sim \mathcal{U}[-0.1, 0.1]$. The disturbances are independent across time.
Sets and sampling.

$$\mathcal{X}_0 = \{x :\; x_1^2 + x_2^2 < 0.01\}, \;\; \mathcal{S} = \{x :\; x_1^2 + x_2^2 \leq 1\}.$$

Initial states are sampled uniformly from $\mathcal{X}_0$, independently of the disturbance sequence.

System 2: valid. The candidate is a polynomial B of degree 10, computed together with SOS multipliers s_0, s_1, s_2 of degree 10 by the SOS program using the method proposed in (Wu et al. 2024):

$$\begin{aligned}\min_{B,\, s_0,\, s_1,\, s_2,\, \gamma} \;\; & \gamma \\ \text{s.t.} \;\; & B,\; s_0,\; s_1,\; s_2 \text{ are SOS}, \\ & B - 1 - s_2\,(\|x\|^2 - 1) \text{ is SOS}, \\ & B - \mathbb{E}[B(x^+) \mid x] + s_1\,(\|x\|^2 - 1) \text{ is SOS}, \\ & \gamma - B + s_0\,(\|x\|^2 - 0.01) \text{ is SOS}.\end{aligned}$$

Each constraint implies one condition. A sum of squares is nonnegative, so $B \geq 0$, which is (2a). For $x \notin \mathcal{S}$, $\|x\|^2 - 1 > 0$ and $s_2 \geq 0$, so $B(x) - 1 \geq s_2(x)(\|x\|^2 - 1) \geq 0$, which is (2b). For $x \in \mathcal{S}$, $\|x\|^2 - 1 \leq 0$ and $s_1 \geq 0$, so $B(x) - \mathbb{E}[B(x^+) \mid x] \geq s_1(x)(1 - \|x\|^2) \geq 0$, which is (2c). We solve the program with YALMIP and MOSEK.

To simulate the changed environments in reality, we actively injected noise b into state x_2 from step $k = 100$ onward:

$$\begin{aligned}x_{2,k+1} = x_{2,k} + 0.01\big(&(0.8 + \xi_k)\, x_{1,k} \\ &+ 10\,(x_{1,k}^2 - 0.21)\, x_{2,k} + b_k\big)\end{aligned} \tag{6}$$

where $b_k \sim \mathcal{U}[0, 0.45]$ and the equation of x_1 is unchanged.

System 2+input: invalid. The candidate of system 2+input is the certificate of System 2. Since both the system and the candidate barrier function are polynomial, we can directly compute the function value at a given point, as well as the $\mathbb{E}B(x^+)$. At $x = (-0.07, 0.30)$ we obtain $\mathbb{E}[B(x^+) \mid x] = 0.203196$ and $B(x) = 0.198623$, so x is a counterexample. Then the candidate in system 2 before is invalid in system 2+input.

System3: polynomial oscillator (2D) (Abate et al. 2021; Vertovec et al. 2025). *Dynamics.* With $x = (x_1, x_2)$ and $\Delta t = 0.1$, the system is

$$\begin{aligned}x_{k+1} &= x_k + \Delta t \begin{bmatrix} x_{2,k} \\ x_{1,k}^3/3 - x_{1,k} - x_{2,k} + u_k \end{bmatrix} + \sigma\xi_k, \\ \xi_k &\sim \mathcal{U}([-1, 1]^2).\end{aligned} \tag{7}$$

The disturbances are independent across time and $\sigma = 0.01$. In the refinement stage, we train the policy and barrier function using neural networks with $|u| \leq 1.5$. We applied Euler discretization to the continuous system from the original text.
Sets and sampling. Let

$$\begin{aligned}\mathcal{X}_0 &= \mathcal{B}((1.5, 0), 0.5) \cup [-1.8, -1.4] \times [-0.1, 0.1] \\ &\quad \cup [-1.4, -1.2] \times [-0.5, 0.1], \\ \mathcal{X} &= [-3.5, 2] \times [-2, 1], \\ \mathcal{D} &= \mathcal{B}((-1, -1), 0.4) \cup ([0.4, 0.6] \times [0.1, 0.5]) \\ &\quad \cup ([0.4, 0.8] \times [0.1, 0.3]).\end{aligned}$$

We define the safe set by $\mathcal{S} = \mathcal{X} \setminus \mathcal{D}$.

System 3: invalid. The policy $u(x) = 1.5\tanh(N_\pi(x))$ and the barrier $B(x) = \mathrm{softplus}(N_B(x))$ are neural networks trained from samples. Finding a valid certificate for such a stochastic system is difficult, and it is not needed here. The closed-loop dynamics are $x^+ = m(x) + 0.01\,\xi$ with $\xi \sim \mathcal{U}([-1, 1]^2)$, where

$$m(x) = \big(x_1 + 0.1\,x_2, \;\; x_2 + 0.1\,(x_1^3/3 - x_1 - x_2 + u(x))\big)$$

is the next state without noise. Since B is a neural network, it's hard to provide the closed form for $\mathbb{E}[B(x^+) \mid x]$. So we bound it by partitioning the noise, as in (Cao et al. 2025), and by interval bound propagation through N_B (Gowal et al. 2019). We split $[-1, 1]^2$ into 20×20 cells C_j of probability $1/400$ each. For $\xi \in C_j$, the successor lies in the box $Z_j =$

$m(x)+0.01\,C_j$, and interval bound propagation through N_B gives a lower bound $\underline{B}_j \le \min_{z\in Z_j} B(z)$. Hence

$$\mathbb{E}[B(x^+) \mid x] \;\ge\; \frac{1}{400}\sum_{j=1}^{400} \underline{B}_j.$$

The upper bound is obtained in the same way. At $x = (0.79, -0.32)$, we obtain $\mathbb{E}[B(x^+) \mid x] \in [0.132798, 0.136322]$ and $B(x) \le 0.126608$. The lower bound of the expectation exceeds $B(x)$, so x is a counterexample and the candidate barrier function is invalid.

Computing environment. The reported experiments were run on Linux with Python 3.10 and NVIDIA A100 40-GB GPUs.

B Further analysis

B.1 Comparison with other monitors

We examine whether history-dependent betting influences the alarm rate over fixed betting rules and how the proposed monitor compares with monitors outside the e-process family. We consider two comparisons. First, we compare our adaptive betting rule with fixed betting rules computed from the same barrier increments. Second, we compare it with CUSUM (Page 1954) whose thresholds are calibrated using safe rollouts. For any betting sequence with $0 \le \lambda_t \le 1/B_t$, define $M_t(\lambda) = \prod_{s=1}^{t}(1+\lambda_{s-1}\delta_s)$. Monitors use the threshold $a^{K_{\max}-t}/\alpha$ of the main experiments ($K_{\max}\ln a = 4$) for their own betting sequences, and they use the average of the $M_s(\lambda)$ values for multiple fixed parameter values when they have the same weights.

The fraction mixture uses $\lambda_t^{(c)} = c/B_t,\ c \in \mathcal{C}$. The absolute mixture uses the fixed absolute bet $\lambda_t^{(\beta)}$ for $\beta \in \mathcal{A}$. In each episode, each component of the absolute mixture keeps c and β fixed, and our method uses the observed increment history to update the betting amount using Eq. (4). CUSUM uses the same increments and alarms when $C_t \ge h$, where C_t is the maximum of $\{0, C_{t-1}+\delta_t\}$ and h is calibrated for safe rollouts. Since the CUSUM method does not provide reliable formal guarantees, we select the parameter h such that its average alarm rate matches that of our method to ensure a fairer comparison.

What do these methods guarantee? For betting components under H_0, their equal-weight mixtures are also nonnegative supermartingales. So Theorem 1 can be applied to Direct, the fraction mixture, and our adaptive monitor. CUSUM relies on thresholds from rollouts, but does not hold for our Theorem 1.

Results. Table 3 and Figure 4 show that all monitors keep the alarm rates of the valid systems below α, with $\hat{p}^{\max} \le 0.50\%$ on System 1 and no alarm on System 2.

On the invalid systems, DAB raises alarm rates above α for 79.90% of the initial states of System 2-input attack and 78.40% of those of System 3. For Direct, these shares are only 17.90% and 23.80%, and for the mixtures they are at most 22.40%. This is because a fixed bet is also placed while the barrier decreases, which multiplies the evidence by a factor below one. For example, the barrier of System 2-input attack decreases for many steps before the attack, and the fixed bets lose evidence during this period. DAB does not bet when the discounted sum S_t of recent increments is negative, and thus keeps its evidence for the steps where the barrier rises. In addition, a mixture raises an alarm only when the average of its components reaches the threshold, so a single growing component must reach several times it. CUSUM, whose threshold is matched to the mean alarm rate of DAB, flags a similar share of initial states but carries no false-alarm guarantee.

Monitor	Bet λ_t	Alarm	Guar.
Direct	$1/B_t$	$M_t \ge \frac{a^{K_{\max}-t}}{\alpha}$	✓
Fraction mix.	$c/B_t,\ c \in \mathcal{C}$	$\bar{M}_t \ge \frac{a^{K_{\max}-t}}{\alpha}$	✓
Absolute mix.	$\min\{\beta, 1/B_t\},\ \beta \in \mathcal{A}_j$	$\bar{M}_t \ge \frac{a^{K_{\max}-t}}{\alpha}$	✓
DAB	$\min\{(S_t/Q_t)_+, U, 1/B_t\}$	$M_t \ge \frac{a^{K_{\max}-t}}{\alpha}$	✓
CUSUM	–	$C_t \ge h$	×

Table 2: Compared monitors, all driven by the increments $\delta_t = B_t - B_{t-1}$. E-process monitors use $M_t = \prod_{s\le t}(1+\lambda_{s-1}\delta_s)$, and mixtures use the equal-weight average $\bar{M}_t$ of their components, with $\mathcal{C} = \{0.2, 0.5, 0.75, 1\}$, $\mathcal{A}_1 = \{1, 10, 20, 100\}$, and $\mathcal{A}_2 = \{10, 30, 60, 100\}$. For Direct, $M_t = B_t/B_0$. In DAB, S_t and Q_t are γ-discounted sums of past δ_s and δ_s^2 (Eq. (4)). CUSUM uses $C_t = \max\{0, C_{t-1}+\delta_t\}$ with an empirically calibrated h. Guar.: $P_{H_0}(\tau \le K_{\max} \mid x_0) \le \alpha$ for every x_0 (Theorem 1).

B.2 Additional Analysis of the Main Experiments

Why is the direct betting $\hat{p}^{\max}$ higher than that of DAB? As shown in Figure 5, the system 1 has lower values in the initial points $B(x_0)$, thereby increasing the probability of an alarm when $B_t/B_0 > 1/\alpha$. In contrast, our method is independent of the initial value and is therefore unaffected by this issue. Overall, these observations are both consistent with the anytime-valid false-alarm guarantee under a valid certificate.

Why does a change in a have different effects on system 2 +input versus system 3? Figure 6(a, b) shows the distribution of DAB's alarm rates over the sweep $K_{\max}\ln a \in \{0, 1, 2, 4, 8\}$: ρ_α stays within 2.2 points on both systems, while $\hat{p}^{\max}$ falls from 92.0% to 81.0% on System 2-input and only from 96.3% to 95.0% on System 3. Figure 6(c, d) shows why. Every threshold equals $\ln(1/\alpha)$ at $k = K_{\max}$, so a run whose evidence is still above $\ln(1/\alpha)$ at the horizon raises an alarm for every a; a larger a can only remove the alarm of a run that crosses $\ln(1/\alpha)$ before $K_{\max}$ and stays below the tilted thresholds. On System 2-input the evidence settles just above $\ln(1/\alpha)$ from about $k = 300$ on, so many runs cross $1/\alpha$ late and only marginally, and the tilted thresholds remove part of them. On System 3 the evidence keeps rising until $K_{\max}$, so almost every run that crosses $1/\alpha$ also crosses the tilted thresholds.

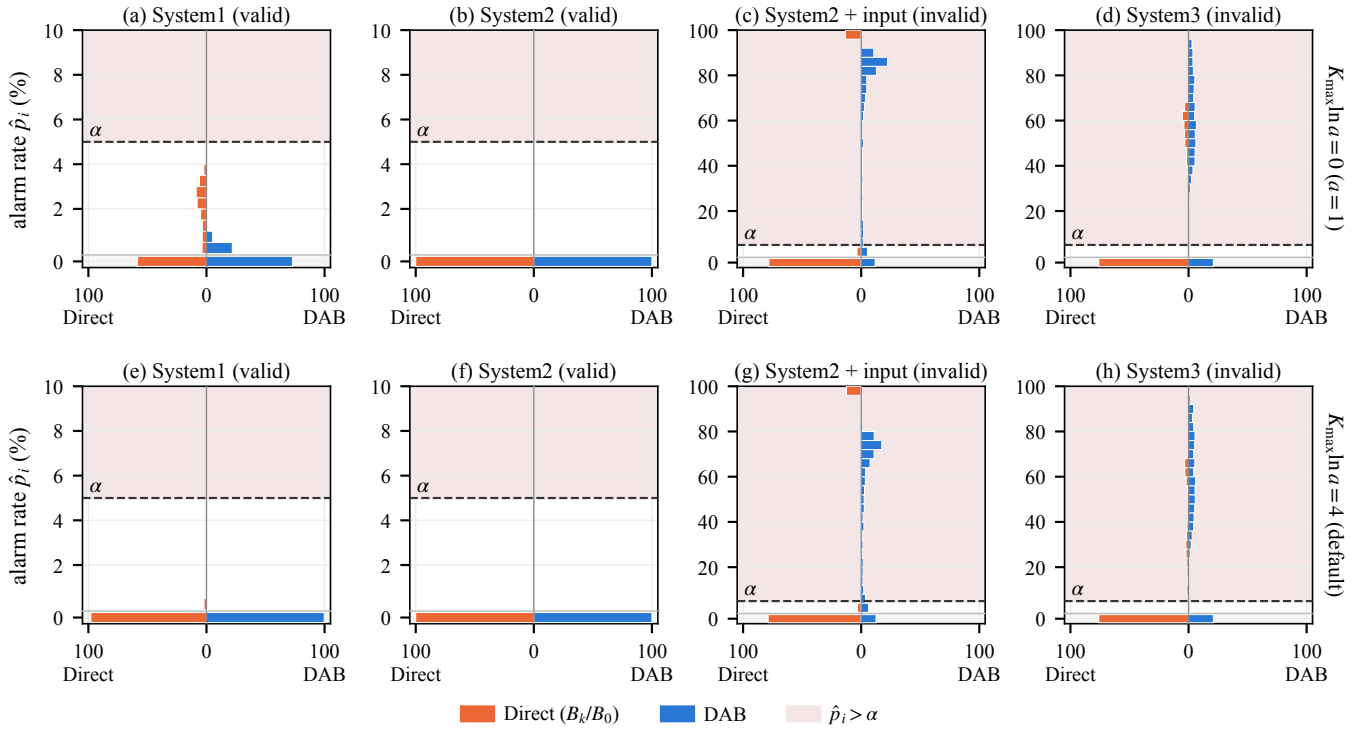


Figure 3: Distribution of the alarm rate $\hat{p}_i$ over the $N = 10^3$ initial states, Direct (left half) against DAB (right half), with the standard rule $K_{\max} \ln a = 0$ (top row) and the default $K_{\max} \ln a = 4$ (bottom row); the row below each axis holds the share of initial states with $\hat{p}_i = 0$.

Why do the two betting strategies differ on the invalid systems? Figure 5 helps explain the different behaviors of the two betting strategies through the distribution of the initial barrier values $B(x_0)$. For direct betting, the evidence process satisfies $M_k = B(\hat{x}_k)/B(x_0)$, so the alarm condition for $a = 1$ is equivalent to

$$B(\hat{x}_k) \geq \frac{B(x_0)}{\alpha}.$$

Thus, a smaller initial barrier value yields a lower alarm threshold, making an alarm easier to trigger. System 2-input attack contains many initial states with small $B(x_0)$, and direct betting reaches a 100% empirical alarm rate for initial states with $B(x_0) = 0.001$. However, this behavior is concentrated among a subset of the sampled initial states, as reflected by $\rho_\alpha = 17.90\%$ (Table 1, $K_{\max} \ln a = 4$; 18.10% with $a = 1$). The same logic applies to the case where $a > 1$, since the threshold $a^{K_{\max}-k} B(x_0)/\alpha$ is still proportional to $B(x_0)$.

In contrast, the initial barrier values in System 3 are generally larger, resulting in higher alarm thresholds for direct betting. Accordingly, Figure 5(b)–(d) shows that its empirical alarm rate decreases rapidly as $B(x_0)$ increases. DAB, however, adapts its betting fraction based on the observed barrier increments and exhibits nonzero empirical alarm rate over a broad range of initial barrier values. This helps explain why DAB achieves a higher fraction of empirically abnormal initial states and a higher worst-case empirical alarm rate than direct betting in System 3.

B.3 Sensitivity analysis

Next, we study how the horizon $K_{\max}$, and the betting rule λ_t affect the alarm rates. In these analyses, we vary only one parameter while keeping the others constant.

Valid certificates. No choice of these parameters can raise the false-alarm probability above α. Under H_0, $\mathbb{E}[\delta_{t+1} \mid \mathcal{F}_t] \leq 0$. The barrier can still increase at single steps, but for every λ_t chosen from past observations, $E_{t+1} \geq 0$ and $\mathbb{E}[E_{t+1} \mid \mathcal{F}_t] = 1 + \lambda_t \mathbb{E}[\delta_{t+1} \mid \mathcal{F}_t] \leq 1$. Hence M_k is a test supermartingale for every γ and U, and Theorem 1 holds for every $K_{\max}$ and every $a \geq 1$. The first two rows of Figure 7 confirm that the alarm rates of System 1 and System 2 stay below α in all settings.

Invalid certificates. In scenarios where certificates are invalid, various parameters will influence the corresponding alarm rates, we will also provide specific recommendations for parameter selection in practical deployment.

1. The horizon factor $K_{\max}$ We vary $K_{\max}$ while fixing $K_{\max} \ln a = 4$, i.e. $a = e^{4/K_{\max}}$, so that the threshold keeps the same shape for every horizon. The run raises an alarm within the horizon $K_{\max}$ if and only if $\tau \leq K_{\max}$. We therefore simulate each run once up to $k = 800$, record the resulting trajectory, and evaluate the alarm rates for different values of $K_{\max}$.

Results. Figure 7(a,d) shows that the alarm rates for the valid systems remain unchanged as $K_{\max}$ varies. Although the alarm rate is non-decreasing in $K_{\max}$ for each fixed set

System	Direct		Fraction mix.		Absolute mix. $\mathcal{A}_1$		Absolute mix. $\mathcal{A}_2$		DAB		CUSUM	
	ρ_α	$\hat{p}^{\max}$	ρ_α	$\hat{p}^{\max}$	ρ_α	$\hat{p}^{\max}$	ρ_α	$\hat{p}^{\max}$	ρ_α	$\hat{p}^{\max}$	ρ_α	$\hat{p}^{\max}$
Valid certificates												
System 1	0.00	0.20	0.00	0.50	0.00	0.00	0.00	0.00	0.00	0.10	0.00	0.10
System 2	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Invalid certificates												
System 2-input attack	17.90	100.00	11.30	100.00	0.00	0.00	0.00	1.20	79.90	81.50	93.80	84.30
System 3	23.80	69.40	0.00	0.00	9.30	29.50	22.40	68.00	78.40	95.10	96.00	99.30

Table 3: Comparison with other monitors on the runs of Table 1. The threshold h of CUSUM is set such that its mean alarm rate equals that of DAB on each system. $N = 10^3$ initial states, $R = 10^3$ trajectories per initial state, $K_{\max} = 400$, $K_{\max} \ln a = 4$, $\alpha = 0.05$. All values are percentages.

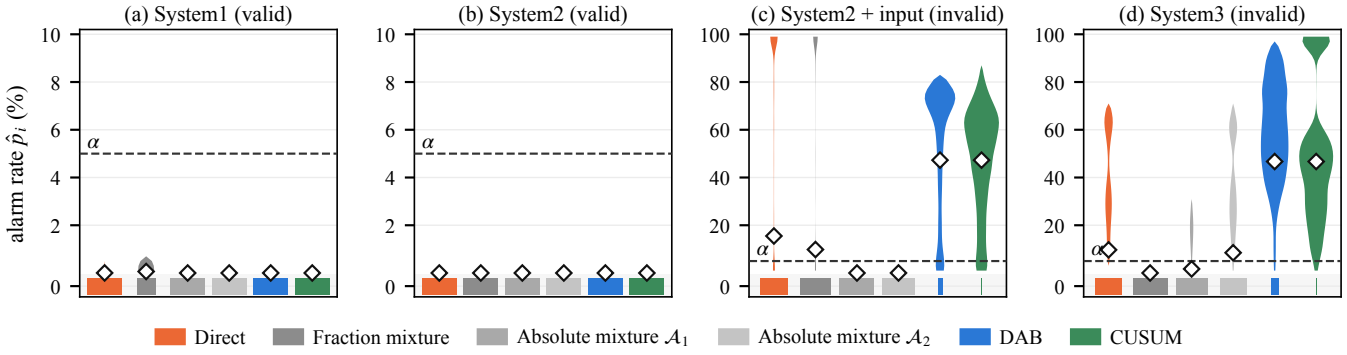


Figure 4: Distribution of the alarm rates $\hat{p}_i$ over the initial states for all monitors ($K_{\max} \ln a = 4$).

of runs, it need not increase strictly; the observed rates remain below or equal to α, consistent with the anytime-valid guarantee. For system 2+input, the attack starts at $k = 100$. Thus, when $K_{\max} = 100$, the monitored trajectory contains no attacked step, and no alarms are raised (Figure 7(g)). For System 3, the barrier function is invalid from the start, but the evidence $\ln M_k$ needs about 300 steps to reach $\ln(1/\alpha)$. At $K_{\max} = 100$, the accumulated evidence $\ln M_k$ remains well below the alarm threshold $\ln(\frac{1}{\alpha})$(Figure 6(d)), so no alarms are raised (Figure 7(j)). For both systems, the alarm rates increase as the horizon grows. From $K_{\max} = 300$ onward, DAB achieves higher alarm rates than Direct betting, indicating that DAB is more effective at detecting the loss of validity of the barrier certificates over these horizons.

In practice, the choice of $K_{\max}$ involves a trade-off between the monitoring horizon and the desired level of confidence in the barrier certificate. A shorter horizon limits the time over which evidence against the certificate can accumulate, whereas a longer horizon provides more opportunities to detect a loss of validity. Since the validity of the barrier certificate is generally unknown a priori, $K_{\max}$ should be selected according to the practical monitoring requirements and the desired level of trust in the certificate. In particular, a shorter horizon may be appropriate when the certificate is considered highly trustworthy and limiting alarms is a priority, whereas a longer horizon may be preferred when more extensive monitoring is needed to detect potential invalidity.

2. Betting rule λ_t. Direct uses $\lambda_t = 1/B(\hat{x}_t)$ and has no tunable parameters. Therefore, its alarm rates remain unchanged across the parameter setting in Figure 7(b, c, e,f). In contrast, DAB uses

$$\lambda_t = \min(\lambda_t^{\mathrm{K}}, U, 1/B(\hat{x}_t)),$$

where λ_t^{K} depends on the discounted sums S_t and Q_t of past increments, with the discount factor γ. When $\gamma = 1$, earlier decreases in the barrier can offset subsequent increases in S_t. As shown in Figure 7(h,k), both invalid systems consequently exhibit substantially fewer alarms. Decreasing γ increases the alarm rate for System 3 but decreases it for System 2+input, indicating that the effectiveness of the discount factor depends on the system.

The cap U mainly limits the betting factor when the Kelly-based factor is not the active constraint. Figure 7(i, l) shows that DAB produces alarm rates above α for System 2+input only for $U \geq 100$, whereas the alarm rates for System 3 change little once $U \geq 13$. These results suggest that the value of U needed to achieve effective detection depends on the system. In practice, U can be selected through a sensitivity analysis, choosing a value beyond which increasing the cap yields little additional improvement in detection performance. Increasing U does not compromise the anytime-valid guarantee, because the constraint $\lambda_t \leq 1/B(\hat{x}_t)$ remains in force. Consistent with this guarantee, the alarm rates for the valid systems remain below α, even when $U = 1000$.

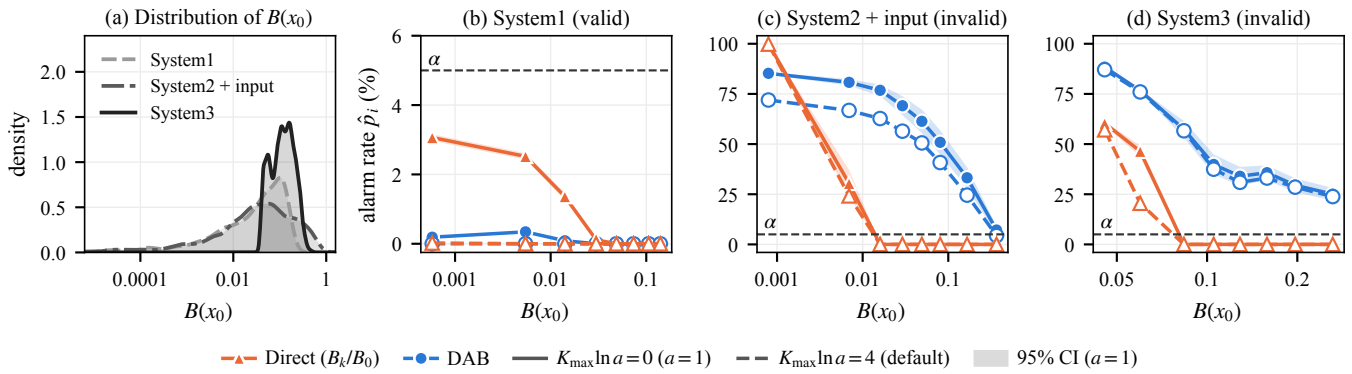


Figure 5: Initial barrier value $B(x_0)$ and alarm rate: solid lines and filled markers for the standard rule $K_{\max} \ln a = 0$, dashed lines and hollow markers for the default $K_{\max} \ln a = 4$.

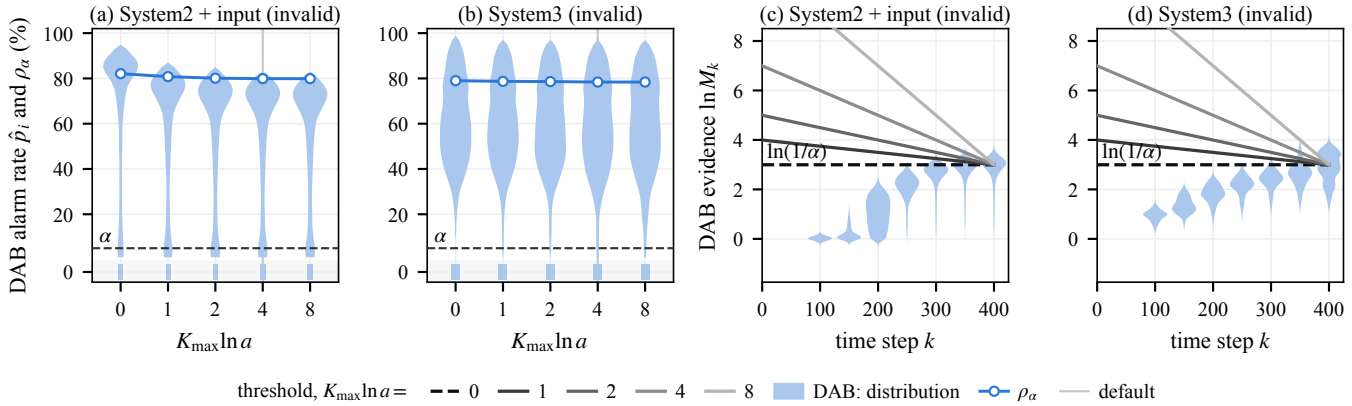


Figure 6: How the patience parameter a impacts alarm rate?

C Training Details

We train the neural barrier B_ϕ (with a softplus output layer so that $B_\phi \geq 0$ structurally) by minimizing soft penalties on the barrier conditions. Given a batch of constraint states $\{x_i\}_{i=1}^N$ sampled from the safe region, we simulate M independent successor states $x'_{i,j} \sim P(\cdot \mid x_i, \pi_\theta(x_i))$ for each x_i and estimate the conditional expectation by $\widehat{\mathbb{E}}[B_\phi(x') \mid x_i] = \frac{1}{M} \sum_{j=1}^M B_\phi(x'_{i,j})$. The supermartingale decrease condition is enforced by

$$\mathcal{L}_{\text{dec}} = \frac{1}{N} \sum_{i=1}^{N} \text{ReLU}\Big(\widehat{\mathbb{E}}[B_\phi(x') \mid x_i] - B_\phi(x_i)\Big), \quad (8)$$

while the initial and unsafe conditions are enforced on sampled sets $\{x_0^{(i)}\}_{i=1}^{N_0} \subset \mathcal{X}_0$ and $\{x_u^{(i)}\}_{i=1}^{N_u} \subset \mathcal{X}_u$ by

$$\begin{aligned} \mathcal{L}_{\text{init}} &= \frac{1}{N_0} \sum_{i=1}^{N_0} \text{ReLU}\big(B_\phi(x_0^{(i)}) - b\big), \\ \mathcal{L}_{\text{unsafe}} &= \frac{1}{N_u} \sum_{i=1}^{N_u} \text{ReLU}\big(1 - B_\phi(x_u^{(i)})\big), \end{aligned} \quad (9)$$

where $b > 0$ is an upper bound on the initial barrier value. The total loss is

$$\mathcal{L}(\phi) = w_1 \mathcal{L}_{\text{dec}} + w_2 \mathcal{L}_{\text{unsafe}} + w_3 \mathcal{L}_{\text{init}}, \quad (10)$$

where $w_1, w_2, w_3 > 0$ are weighting parameters. All terms are optimized with Adam. In this paper, we mainly focus on online training, where collecting multiple independent successors from the same state may be impractical. We therefore set $M = 1$. Although the resulting single-sample Monte Carlo estimate is unbiased for $\mathbb{E}[B_\phi(x') \mid x_i]$, it has higher variance. Moreover, applying the nonlinear ReLU penalty to a single-sample estimate does not provide an unbiased estimate of the ideal expected decrease loss. Consequently, the learned barrier is not necessarily a valid certificate. Nevertheless, the runtime monitor can detect potential violations and intervene against unsafe behavior. The neural network output B_ϕ uses a Softplus activation, which ensures that $B_\phi(x) \geq 0$. We then define the deployed barrier function by the piecewise extension

$$\bar{B}_\phi(x) = \begin{cases} B_\phi(x), & x \in \mathcal{S}, \\ 1, & x \notin \mathcal{S}. \end{cases} \quad (11)$$

Therefore, the nonnegativity condition and the unsafe-set condition $\bar{B}_\phi(x) \geq 1$ for all $x \notin \mathcal{S}$ are satisfied structurally.

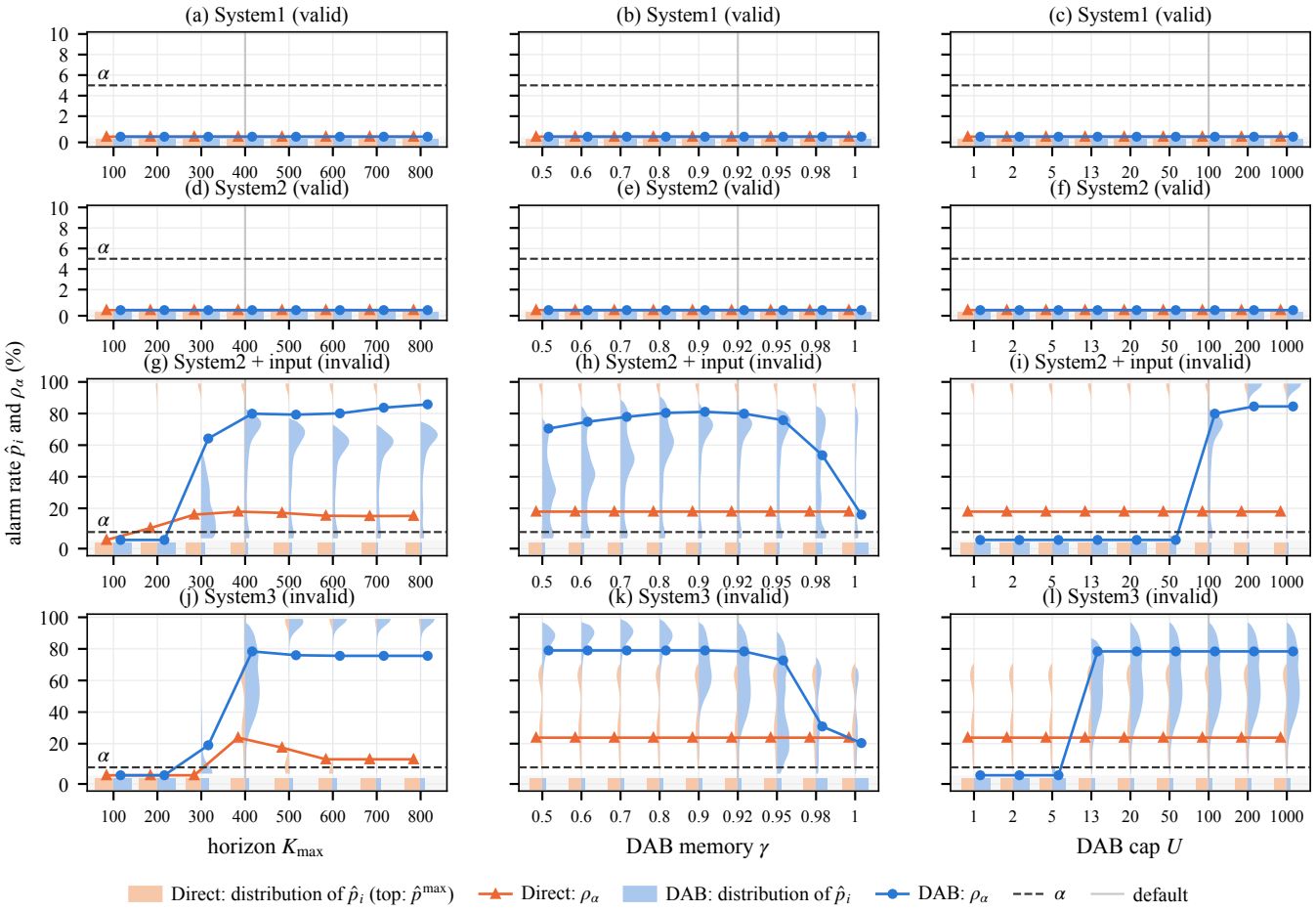


Figure 7: Sensitivity to the horizon $K_{\max}$, DAB memory γ and DAB cap U.

During deployment, the runtime monitor evaluates $\bar{B}_\phi$ and computes the barrier increments as

$$\delta_{t+1} = \bar{B}_\phi(\hat{x}_{t+1}) - \bar{B}_\phi(\hat{x}_t). \tag{12}$$

The sampled unsafe-set loss is retained as an auxiliary training term to encourage the neural-network output to approach the prescribed unsafe value near the boundary of the safe set.